
Lawyers & Coders: Development and Decisions

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1. Introduction

“Do you also think nature needs a good lawyer?”

This was the question that immediately drew us to this project. Our answer was simple: *yes, we do*. The next question was: *how can we help?*

In February 2026, the second graduating cohort of the Bachelor of Science in Computational Social Science at the University of Amsterdam was presented with a selection of Capstone projects. Among them was a project focused on advancing environmental justice through legal action, presented by a collaboration between *Advocaat van de Aarde* and 010 Coding Collective.

Advocaat van de Aarde is an independent, volunteer-based legal organisation that supports citizens and civil society in defending their right to a healthy environment. Through strategic litigation and legal expertise, it works to hold polluters accountable and strengthen environmental protection through the enforcement of law (<https://advocaatvandeearde.nl>).

010 Coding Collective is a Rotterdam-based software and AI development collective providing technical expertise for organisations working with AI systems. They support projects ranging from refining AI-generated applications to providing development support and strategic advice. As collaborators with *Advocaat van de Aarde*, they contribute technical expertise in the collection, processing, and use of environmental data (<https://010codingcollective.com/en/>).

The challenge they presented was clear. Although large amounts of environmental data are publicly available, this information is often scattered across different sources, stored in incompatible formats, and difficult to interpret without technical expertise. As a result, valuable information that could support environmental litigation remains difficult to access and use effectively.

Could a team of students build a tool that brings these datasets together and helps environmental lawyers build stronger cases to protect the environment?

This question led to the formation of *Lawyers & Coders*. Our team, Melody, Khushi, Aya, Yuchia, and Davide, set out to explore how computational tools could support environmental litigation and improve access to environmental information.

This document forms part of the Tool Manual for the mapping platform developed during our Capstone project. It explains the decisions, challenges, and development process behind the tool, providing insight into how the project evolved from an initial idea into a working platform. Our aim is not only to document what we built, but also to reflect on why certain choices were made, what limitations remain, and what future improvements may be possible.

Over the course of this project, we learned that improving access to environmental information is an ongoing process rather than a finished task. As long as information exists, there will always be opportunities to make it more accessible, usable, and actionable.

This mapping platform is therefore not an endpoint, but a starting point.

2. Acknowledgements

This project would not have been possible without the support of several people, and we want to take this opportunity to thank them properly.

A special thanks goes to Ivonne from Advocaat van de Aarde, who was an exceptional project partner from start to finish. She was always reachable, made time for meetings, communicated the project's needs with clarity, and offered consistent encouragement throughout. Her trust in us gave us the confidence to see this through.

We also want to thank Jantine and Max, also from Advocaat van de Aarde. Jantine and Max provided valuable feedback on the tool and gave us deeper insight into the practical needs of the project, as well as connecting us with more interviewees. We are grateful for the time and dedication both of them gave.

Thank you to Regina Nockerts, our core lecturer and supervisor, who met with us every week, answered our questions no matter how small, and consistently pushed us to aim higher. Her guidance shaped this project in more ways than we can count.

Finally, thank you to 010 Coding Collective for their technical support. They responded to questions promptly via Trello and dedicated time to meeting with us, providing the technical expertise that made the prototype possible.

Lawyers & Coders (Melody, Khushi, Aya, Yuchia & Davide)

3. Project Definition

Our Interpretation of the Project

As a Capstone project, we needed to balance the practical needs of the project partners with our own academic research objectives. While the project description clearly outlined the need for a digital intervention, our goal was not only to build a technical solution but also to investigate the broader structural challenges that make environmental data difficult to use for legal action. This meant continuously translating stakeholder needs into both research

questions and design requirements.

Initial Understanding

The project description focused on improving access to fragmented environmental data for environmental litigation. The primary stakeholder was Advocaat van de Aarde. Our initial concern was that the scope appeared relatively narrow and focused on specific data sets.

Reframing the Problem

As we engaged with the literature and theoretical frameworks, we realised that the project addressed a broader challenge: *Public environmental data exists, but is often difficult to access, interpret, combine, and use in practice.*

How We Interpreted the Project

Rather than just developing a fully finished product, we approached the project as an exploration of:

- How access to information influences environmental justice
- How environmental litigation is a form of collective action
- How fragmented environmental data can be integrated.
- How data can be translated into evidence for legal action.
- How digital tools can support environmental governance and legal action.

Final Understanding

The tool was not intended to determine whether environmental violations had occurred, nor to replace scientific or legal expertise.

Instead, its purpose was to:

1. Improve access to fragmented environmental information.
2. Allow non-technical users to interpret complex environmental data.
3. Support the identification of potential environmental harms.
4. Enable the creation of transparent and reproducible evidence.
5. Help environmental lawyers and NGOs in preparing and strengthening legal cases.
6. Support environmental participation, accountability, and justice.

4. Problem Exploration

What Problems Were Identified

During the early stages, and throughout the development, of the project, we conducted many interviews, both formal and informal ones, with a wide range of stakeholders to better understand the challenges surrounding environmental data and accountability in the Netherlands.

These stakeholders included:

- Environmental Lawyers
- Policymakers
- Environmental scientists
- GIS specialists / data experts
- Environmental NGOs
- Journalists

The interviews helped us better understand the technical and systemic barriers surrounding environmental data, as well as the workflows of the people who would potentially use the platform. These findings helped us identify both the main audiences for the platform and the functionalities that would be most important to prioritise.

There were several recurring issues brought up:

- Environmental data is technically public but difficult to access and interpret for non-experts
- Data is fragmented and disconnected
- Environmental investigations require significant manual work and specialist expertise
- Data often lacks clear traceability, metadata or legal context
- Data needs to be traceable and linked to legal standards to be used in court
- Existing GIS systems are often too technical or inaccessible for non-specialists

Summary of Expert Inputs

	<i>Challenges They Face</i>	<i>Their Use Cases for the Tool</i>	<i>Their Needs</i>
Environmental Lawyers	- Linking data to legal thresholds and permits - Depending on costly specialist expertise - Courts strongly trust official monitoring systems	- Investigating environmental violations - Supporting litigation and permit challenges - Exploring spatial relationships between pollution and sensitive areas	- Traceable & reproducible data - Exportable evidence - Clear links between datasets and legal standards - Easy navigation
Policymakers	- Fragmentation across governance levels - Difficulty communicating environmental uncertainty to the public - Maintaining reliable and standardized monitoring systems	- Public transparency and communication - Environmental monitoring and coordination - Exploring environmental trends spatially	- Reliable and updated data - Clear metadata and methodological explanations - User-friendly interface
Environmental scientists	- Environmental systems are highly complex	- Visualizing environmental patterns	- Scientific disclaimers and

	- and context-dependent - Monitoring data may underestimate actual environmental harm - Fixed spatial assumptions can oversimplify scientific reality	- and trends - Supporting environmental analysis - Identifying areas for further investigation	- contextualization - Visualizations - Flexible spatial analysis tools
GIS & Data Specialists	- Inconsistent file formats, scales, and coordinate systems - Outdated datasets and metadata issues - Difficulty discovering authoritative datasets	- Combining and analyzing spatial datasets - Environmental risk mapping - Interactive spatial data	- Fast performance with large datasets - Consistent coordinate systems - Flexible and scalable backend architecture
Environmental NGOs	- Large amount of manual work needed for investigations - Dependence on technical experts - Difficulty accessing and interpreting public datasets	- Strategic litigation - Pollution hotspot identification - Public awareness and advocacy	- Integrated datasets - User-friendly interface - Exportable maps and reports - Transparent data provenance
Journalists	- Environmental data spread across many institutions - Significant manual verification required - Difficulty translating technical information into public stories	- Data-driven investigations - Spatial storytelling and visualization - Identifying environmental anomalies	- Strong visualization tools - Clear source transparency

Key Insights and Design Priorities

The interviews showed that the biggest issue was not the absence of environmental data, but the difficulty of accessing, combining, and understanding it. Stakeholders repeatedly described environmental investigations as slow and highly dependent on technical expertise.

Another major insight was the importance of transparency and traceability. Many stakeholders emphasized that environmental data becomes far more useful when users can clearly see where it came from, how it was processed, and how it connects to legal or environmental standards. They also highlighted the need to balance accessibility with scientific caution. While the platform should simplify environmental information for non-technical users, it also needs to communicate uncertainty, limitations, and contextual information clearly.

Based on these findings, we identified several core priorities for the platform:

- Integrating fragmented environmental datasets into one interface
- Simplifying spatial analysis for non-technical users
- Improving transparency through metadata and source traceability

- Supporting exportable evidence and reproducible workflows
- Prioritizing fast and interactive map visualization
- Designing a flexible backend capable of handling inconsistent datasets

5. Requirements Development

The platform functionalities and content were developed iteratively throughout the project based on discussions and feedback sessions with our project partners, the 010 Coding Collective, meetings with our Core Lecturer Regina, focus group and prototype event feedback, stakeholder interviews, as well as our own continuous experimentation and trial-and-error process.

Our initial requirements were largely defined by our project partners, who outlined the core functionalities and datasets most relevant to their environmental investigations and legal workflows. From there, the platform expanded significantly through expert interviews and prototype testing. As development progressed, requirements continued evolving based on stakeholder feedback, technical limitations, and usability concerns.

An Interactive Web-Based Map

The platform needed to function as an interactive online GIS interface where users could visually explore multiple environmental datasets together in one place.

Core Platform Requirements

Map Layer Management	Users needed the ability to: - Switch layers on and off - Adjust transparency - Combine and filter map layers - Compare datasets spatially - Create new layers → one of the most important requirements because stakeholders repeatedly described the difficulty of manually comparing disconnected datasets across different systems.
Spatial Selection & Buffer Analysis	The platform needed tools for: - Clicking and selecting map objects - Selecting areas manually - Drawing configurable buffers - Exploring spatial relationships between datasets → important for environmental investigations and identifying potential risk zones near sensitive locations such as Natura 2000 areas, schools, or residential buildings.

Dataset Integration	A major requirement was integrating multiple fragmented datasets into one interface, including: - BRP crop parcels - Cadastral data - Natura 2000 areas - BAG buildings - Livestock emission data - Schools & healthcare locations - Pesticide measurements Later on in the process, based on more rounds of feedback (especially from our second round of Focus Group), we added additional datasets consisting: - The Natuurnetwerk Nederland (NNN) - Administrative boundaries - Water Authorities Hydrography - Water Framework Directive (WFD) surface water body boundaries - Water authorities administrative boundaries
Data Export & Evidence Generation	- Export filtered datasets to Excel/CSV - Export map views as PDFs/images - Export source references and metadata → important for NGOs, journalists, and legal professionals building reports or legal cases
Data Transparency & Traceability	Ensuring transparent data provenance so that users can clearly see: - Where datasets came from - When datasets were updated - What methodology or source was used → important for legal credibility and reproducibility
Timeframe & Historical Data	Several datasets, particularly BRP crop parcels, required historical filtering and year selection. This led to the addition of: - Year sliders - Historical layer selection - Time-based filtering

Other points that were also emphasized were:

- User-friendliness to ensure that the platform was understandable for non-technical users. This included:
 - Simple user interface design
 - Clear visualizations
 - Information boxes and walk-throughs

- Minimal GIS knowledge required
- Flexible architecture that accommodates to frequently-changing environmental datasets, which would allow (essentially, futureproofing):
 - New datasets to be added later
 - Backend that remains modular and scalable rather than hardcoded around fixed datasets

Datasets and Their Specifications

Dataset	Description	Source	Specifications
The Basisregistratie Gewaspercelen (BRP) link	Registers all agricultural parcels and their declared crops across the Netherlands each year. Used for EU subsidy administration and agricultural policy monitoring.	PDOK	- Colour-coding parcels based on crop type - Historical yearly comparisons - Grouping crops into broader categories (e.g. grains) - Surface area calculations - Summary tables showing crop changes within selected regions - Distance calculations between parcels and sensitive locations
The Basisregistraties Adressen en Gebouwen (BAG) link	The national register of all addresses and buildings in the Netherlands, including construction year, usage type, and official status.	PDOK	- Display of usage locations/buildings - Relationship with surrounding parcels and areas - Counting of residential/usage locations within a defined buffer
Natura 2000 Areas link	EU-designated protected nature areas under the Birds and Habitats Directives. Activities within or near these boundaries are subject to strict environmental permit requirements.	PDOK	- Polygon-based buffers with configurable distances - Clear legends and center-point calculations - Links to governmental explanations and legal context surrounding Natura 2000 protections

The Natuurnetwerk Nederland (NNN) link	A national network of nature areas aimed at protecting and connecting biodiversity, defined per province under the Dutch Nature Protection Act.	Nationaal Georegister	→ Added as additional dataset to expand on Natura 2000 areas
Cadastral Data link	Official cadastral parcel boundaries and sizes across the Netherlands, which is essential for identifying land ownership in legal and environmental disputes.	Nationaal Georegister	- Display Cadastral Parcels - Basic Information Attributes - Linking selected BRP crop parcels to cadastral references - Data exportation
KRD Veehouderijen link	Registered livestock farms in the Netherlands with their NH3 (ammonia), odour, and particulate matter emission values. Used in legal assessments of cumulative farm impact on nature areas.	Kernregistratie Diervverblijven	- Emission figures for livestock facilities
Bestrijdingsmiddelen Atlas link	Pesticide concentration measurements in Dutch surface water from the Bestrijdingsmiddelenatlas, showing which substances exceed environmental quality standards and by how much.	Bestrijdingsmiddelen Atlas	- Relevant measurement data - Linking agricultural parcels with detected pesticides - Closest pesticide point visualisation (toggle)
Health Facilities link	Locations of hospitals, clinics, pharmacies, and other health facilities in the Netherlands.	OSM	- Points showing locations of healthcare facilities
School Institutions link	All registered educational institutions in the Netherlands, from primary schools (basisonderwijs) to universities.	Dienst Uitvoering Onderwijs (DUO)	- Points showing locations of schools
Bestuurlijke Gebieden link	Administrative boundaries consist of municipal, provincial, and national borders, compiled based on the cadastral register (BRK). This dataset contains the history.	PDOK	→ Added as additional dataset

Water Authorities Hydrography link	INSPIRE-harmonised watercourse data from Dutch water authorities (waterschappen), covering rivers, canals, and drainage channels with stream order and condition attributes.	PDOK	→ Added as additional dataset
WFD Surface Water link	Water Framework Directive (WFD) surface water body boundaries covering rivers, lakes, and coastal waters assessed for ecological and chemical status under EU law.	PDOK	→ Added as additional dataset
Waterschappen link	Administrative boundary areas of the 21 Dutch water authorities (waterschappen), responsible for water management, flood protection, and water quality.	PDOK	→ Added as additional dataset

Feedback sessions significantly shaped the evolution of the platform. Several datasets were added or expanded after stakeholders explained their importance in real legal and environmental workflows, particularly the NNN layer, water governance datasets, and BAG.

Other changes focused more heavily on usability and accessibility. Early prototypes were more technically oriented, but feedback repeatedly emphasized the need for simpler terminology, clearer legends, better rendering communication, and stronger contextual explanations for non-experts (refer to *Section 7, UI / UX Iterations* for further details).

6. Design Decisions

Mapping Library Choice

Through a combination of desk research, a meeting with a GIS expert, and input from a technical expert, we developed a shortlist of potential mapping approaches. The table below summarises the main advantages and disadvantages of each option, followed by a justification of our final choice.

Mapping Approach	Advantages	Disadvantages
<i>Option 1:</i> Leaflet + GeoJson	- Simple, fast-to-build web mapping solution - Very good integration	- Limited performance with larger datasets - No built-in spatial

	with GeoJSON - Debugging and maintenance are easy - Large ecosystem of plugins for interactivity	analysis capabilities - 3D visualisation not supported natively - Requires additional tools if analytical logic is needed
<i>Option 2:</i> GeoJSON + turf.js + deck.gl	- Most flexible and powerful web-based solution - Turf.js enables in-browser spatial analysis (buffers, overlays, intersections) - Deck.gl provides high-performance rendering - Scales well with large datasets (especially compared to Leaflet) - Supports advanced visualization (including 3D and layered data)	- Highest technical complexity and learning curve - Requires careful data architecture design - Debugging and maintenance more difficult
<i>Option 3:</i> QGIS-LTR	- Very strong spatial analysis capabilities - No licensing cost - Excellent for data cleaning, preprocessing, and spatial modelling - Stable version, suitable for long-term projects - Good for preparing datasets for web deployment	- Not web-based (requires export to use in final product) - Limited real-time interactivity for end users - Collaboration is difficult - Requires separate deployment pipeline for web integration
<i>Option 4:</i> ArcGIS Pro	- Very powerful and professional GIS environment - Strong integration with ArcGIS Online and web services - Great for collaboration - User-friendly compared to many GIS alternatives - Strong support and documentation	- Paid software with licensing constraints (though accessible as University of Amsterdam students) - Access may be lost after graduation - Closed ecosystem (cannot host separately from ecosystem) - Less flexible for custom features

Key Comparisons

Mapping Approach	Ease of Use	Analytical Power	Web Integration	Performance	Flexibility
Option 1: Leaflet + GeoJson	High	Low	High	Medium	Medium
Option 2: GeoJSON + turf.js + deck.gl	Low	Very High	Very High	Very High	Very High
Option 3: QGIS-LTR	Medium	Very High	Low	High	Medium
Option 4: ArcGIS Pro	Medium	Very High	Medium (via ecosystem)	High	Low

Final Decision

From first glance, **Option 2 (GeoJSON + turf.js + deck.gl)** seemed like the best choice because it scored highly in our evaluation. However, it would have required us to learn and combine several different tools within a short Capstone timeframe. We felt that the extra time needed to understand and implement these technologies could take away from developing the core features of the project.

Instead, we chose **Option 1 (Leaflet + GeoJSON)** because it was simpler, reliable, and easier to work with. It provided all the functionality we needed for the interactive map while allowing us to focus on building the tool, integrating datasets, and improving the user experience. Given the limited time available, it was the most practical choice for the project.

7. UI Iterations

Basic Wireframe Development

- Phase 1: Initial Prototype & Low-Fidelity Wireframe

We began with simple pen-and-paper (digitally) sketches to explore what the map interface could look like. These early ideas were presented for feedback to identify gaps and missing elements. Based on this input, we refined the concept and moved on to a more developed version.

DRAFTS

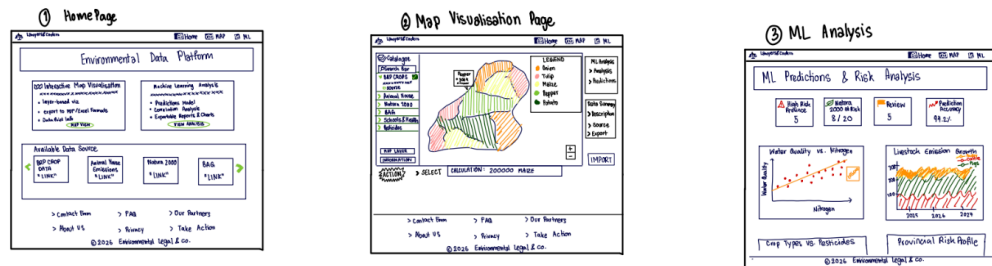


Figure 1: Overview of low-fidelity wireframe

By this stage, we had developed a clear understanding of the project's requirements and the functionalities the tool needed to include. Before beginning development, we wanted to explore the best ways to visualise and implement these features. To do this, we created a series of sketches outlining the homepage, the interactive map interface, and the machine learning analysis page. These early designs helped us think through the user experience, page layout, and key functionalities before moving on to the coding phase.

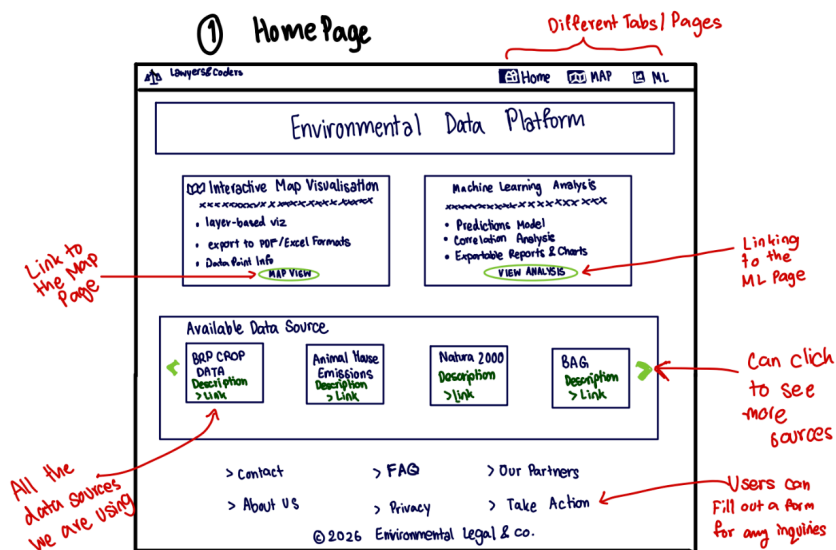


Figure 2: Homepage design (low-fidelity wireframe)

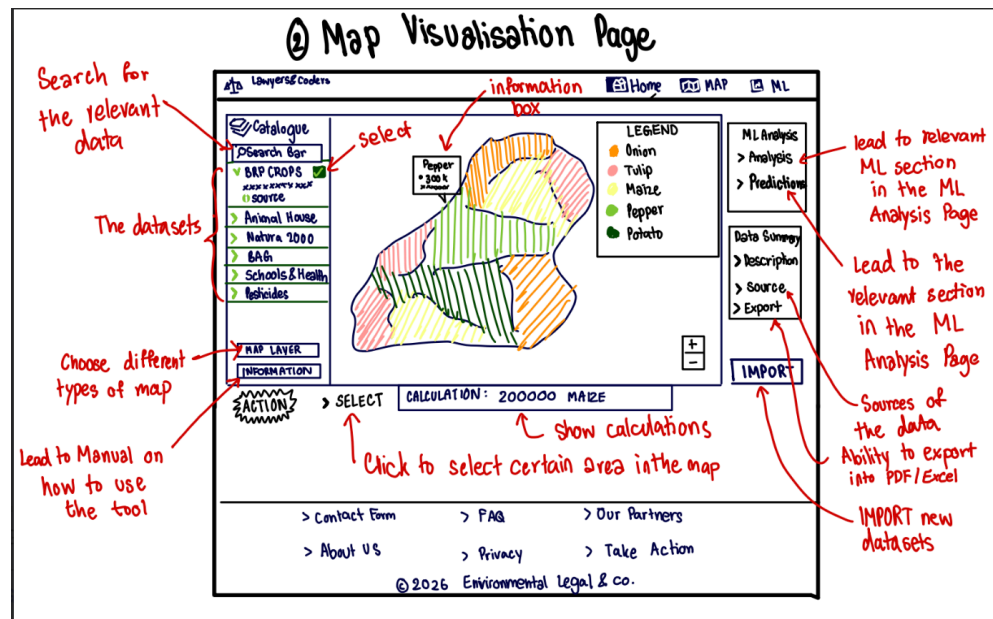


Figure 3: Map page design (low-fidelity wireframe)

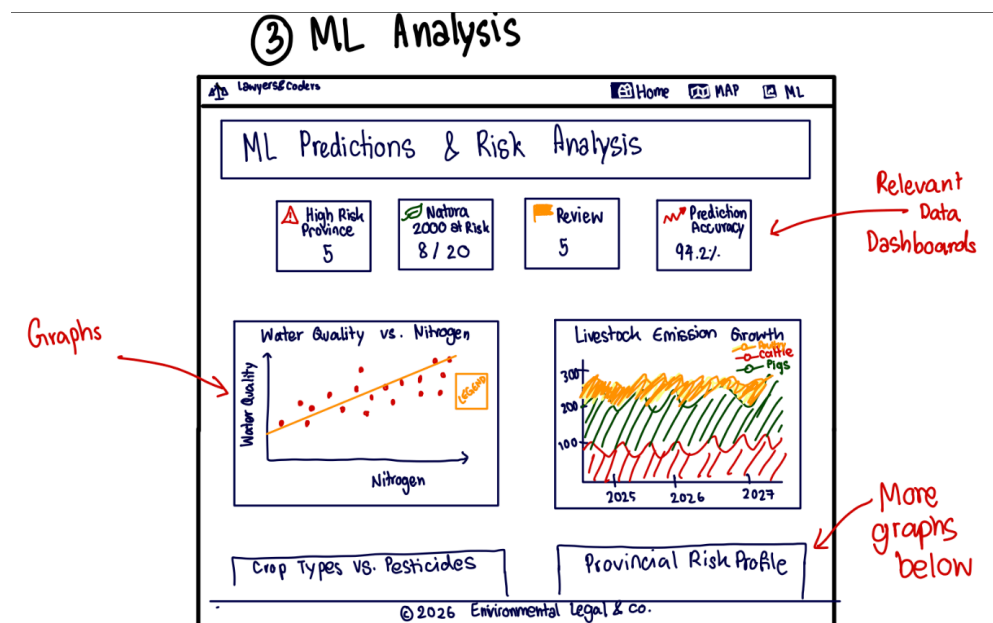


Figure 4: ML Analysis page design (low-fidelity wireframe)

- Phase 2: High-Fidelity Wireframe

Using Lovable, we created a high-fidelity wireframe that not only visualised the design but also helped clarify the logic behind buttons and key functions. This step was important because it allowed us to understand what code and data flows would be needed to make the website work properly. The prototype was then tested during the mid-term prototype event, where participants interacted with it for the first time and provided feedback.

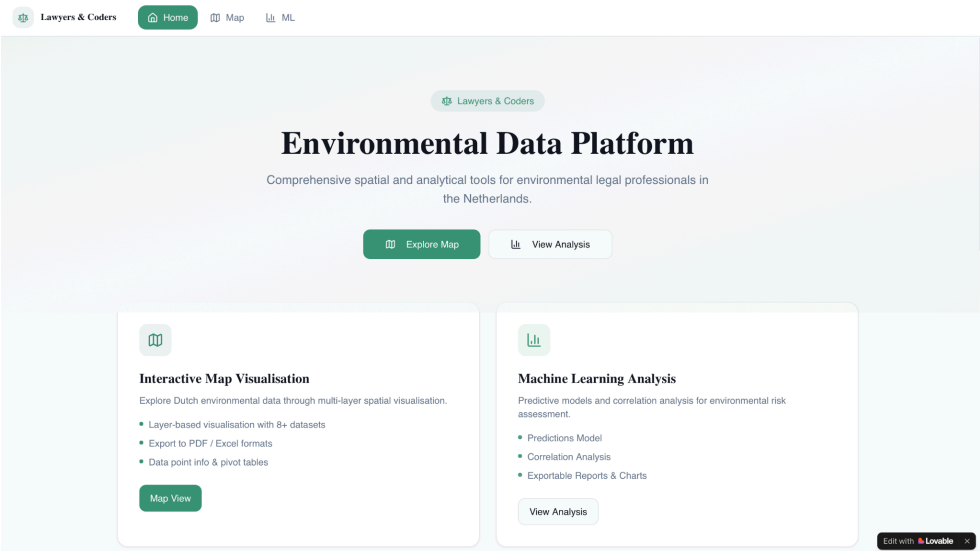


Figure 5: Homepage design (High-fidelity wireframe)

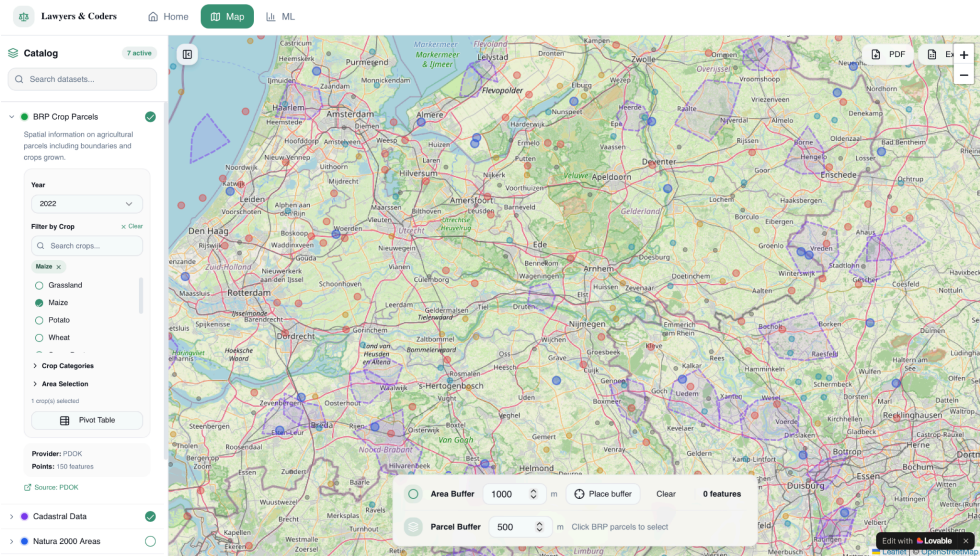


Figure 6: Map page design (High-fidelity wireframe)

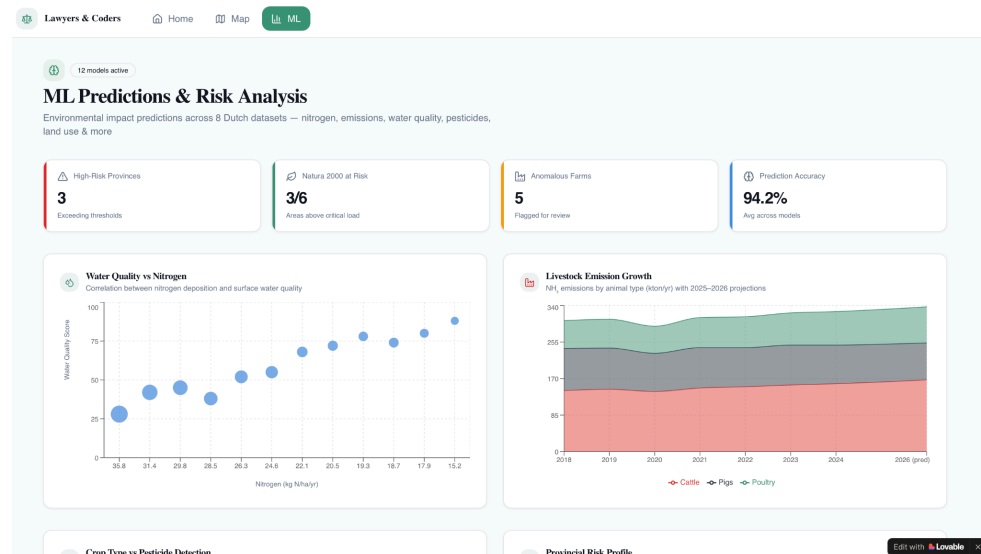


Figure 7: ML Analysis page design (High-fidelity wireframe)

The feedback highlighted several key improvements:

Category	Feedback	Changes
<i>Color Scheme</i>	Colours were too similar	Adjust the colour palette to improve contrast and clarity
<i>Dataset Information</i>	The dataset was not clearly explained	Added basic dataset information, an "info" (i) icon for additional context, and direct links to original data sources.
<i>Website Purpose</i>	The purpose of the website was unclear	Added a short, impactful one-sentence introduction on the homepage, supported by additional contextual information.

- Phase 3: Iterative Prototype Development

Through several focus groups, we continued refining the prototype. Each iteration introduced improvements based on user feedback, gradually shaping the final version of the tool. The feedback and corresponding changes are summarised below.

Category	Feedback	Changes
<i>User Interface</i>	Interface was perceived as professional, intuitive, and visually strong	Visual design largely kept as-is, only minor refinements expected
<i>Data Layers</i>	Woondeals layer incorrect (not actual administrative boundaries)	Replace with PDOK Bestuurlijke Gebieden dataset
	Missing NNN (Natuur Netwerk Nederland) layer	Add as key environmental/legal dataset
	Missing cadastral parcel numbers	Required for legal document generation and must be added
	BAG dataset missing verblijfslocaties sublayer	Add to enable exposure analysis near residences
	Missing water board boundaries	Add due to frequent use in litigation
	Broken or incorrectly implemented datasets identified during testing	Improve dataset validation and domain review process
	No breakdown of sub-crop composition	Add more granular crop-level data
<i>Export Functionality</i>	PDF screenshot export and Excel export positively received	Keep export features, but improve analytical depth
	Missing land-use categories (e.g. corn vs grassland)	Add land-use filtering option and include results in exports
	No distance-to-feature calculations (e.g. schools, Natura 2000)	Add distance calculations
<i>Buffer Analysis Tool</i>	Strongly positively received (high practical value)	Retain and further develop →
		Add: counts of features within buffer
		Add: total area calculations
		Add: distance summaries
<i>Language & Accessibility</i>	Concern about	Consider Google translate

	English-only interface for Dutch legal context	API integration for translation
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8. Key Challenges

Throughout development, the project encountered several technical and practical challenges related to environmental data integration, geospatial processing, scalability, and usability. Many of these issues emerged directly from the fragmented nature of Dutch environmental datasets and the limitations of working with large-scale spatial data in a web environment.

Data Fragmentation

1. *Different File Types*

One of the first major challenges was that every dataset came in a completely different format or had to be accessed differently. Some were file-based GeoJSON, CSV, Shapefile (.shp), or GeoPackage (.gpkg). Others were only accessible via live web services: WMS (Web Map Service) for pre-rendered map tiles, WFS (Web Feature Service) for raw vector features, or PDOK's OGC API for paginated feature requests. Each combination required a completely different loading and parsing approach in the ETL scripts.

2. *Different Coordinate Systems (EPSG)*

Coordinate systems became a major recurring issue throughout development. Most Dutch governmental datasets use RD New (EPSG:28992), while the web map itself operates in WGS84 (EPSG:4326). As a result, every dataset had to be reprojected before it could be visualized correctly on the map.

Some datasets also lacked proper CRS metadata entirely. For example, the healthcare dataset imported from HOTOSM did not specify a coordinate system, forcing the team to make assumptions during processing. Incorrect reprojection could result in features appearing in the wrong location without immediately obvious errors.

3. *Different Dates and Schema Structures*

Several datasets changed structure across different years. The BRP crop parcel dataset was especially difficult because data is published separately per year, crop categories and field names change over time, and historical filtering had to be supported in the frontend. This required dedicated ingestion scripts, schema migration handling, and year-based filtering logic to allow users to compare crop parcels historically.

4. *Different Geometry Types*

The project combined different geometry types, each of which required different rendering strategies, interaction behaviors, legends, and spatial analysis methods within the frontend.

Geometry Types	Points (schools, hospitals, livestock farms)
	Polygons (crop parcels, Natura 2000 areas)
	Administrative boundaries
	Waterway geometries

Performance with Large Datasets

BRP Crop Parcels

The BRP crop parcel dataset covers the entire Netherlands across multiple years and contains extremely large polygon datasets with many crop categories. Loading the data required chunked database inserts (50,000 rows at a time), increased database timeouts, simplified geometries, and optimized rendering strategies. Without these optimizations, the ingestion process would regularly crash or timeout.

Cadastral Data

Cadastral parcel data was even larger, consisting of millions of parcel geometries across the Netherlands. Rendering raw cadastral polygons directly in the browser proved impractical due to performance limitations. As a result, the project switched to using PDOK WMS tile layers instead of rendering the raw geometries client-side.

Frontend Rendering & Map Lag

Large spatial datasets created noticeable lag and rendering delays within the frontend. The frontend had to adopt `preferCanvas: true` in Leaflet (noted in the code as "crucial for rendering thousands of local polygons"), geometry simplification via `turf.simplify()`, and server-side simplification. Natura2000 and NNN layers were also switched to WMS tiles for the same reason. Despite these optimisations, users still experienced noticeable load times when toggling layers.

API and Web Service Limitations

1. *Slow API Loading* - Some datasets relied on paginated APIs with strict request limits. For example, the hydrography dataset from PDOK only returned 1,000 features per request, requiring full pagination loops to load the complete dataset. This significantly slowed ingestion and required custom retry logic with exponential backoff to handle failed requests.

2. *Risk of External API Changes* - The project relied heavily on external governmental APIs and services such as PDOK. This created long-term maintainability concerns because endpoints occasionally change. Additionally, APIs sometimes return unexpected formats and scale limitations can return XML errors instead of JSON responses. As a result, changes on the provider side could break parts of the platform.
3. *Datasets Without APIs* - Some important datasets were not available through live APIs at all. These datasets had to be manually downloaded, stored locally, and periodically re-imported, making automation and maintainability more difficult.

Spatial Accuracy Issues

1. *Reprojection Accuracy* - Because every dataset used different coordinate systems, reprojection errors became an ongoing concern. Incorrect or missing CRS definitions could shift geometries into incorrect locations on the map. This became especially important for legal and environmental investigations where spatial precision matters.
2. *Geocoding Address-Based Data* - School datasets originally contained only addresses rather than coordinates. These addresses had to be geocoded into latitude and longitude values. This introduced additional uncertainty because geocoders approximate locations based on textual matching rather than exact survey coordinates. Schools with ambiguous or incomplete addresses could therefore appear slightly offset on the map. The geocoding workflow went through multiple iterations before stable coordinates were achieved.

UI/UX & Rendering Challenges

1. Rendering Limitations

Certain layers are only rendered correctly at specific zoom levels due to feature density and browser performance limitations. This created confusion for users when datasets appeared to disappear while zooming.

To address this, the interface eventually incorporated:

- Rendering warnings
- Zoom-level guidance
- Loading indicators
- Information pop-ups explaining rendering behavior

2. Buffer Tool Complexity

The buffer analysis tool became one of the most valuable features, but also one of the most technically difficult to refine. Challenges included correct center-point calculations, polygon highlighting behavior, rendering large buffers, and maintaining performance while running spatial queries.

3. *Information Overload*

Combining many datasets into one interface created risks of cluttered maps and overwhelming information density. Several iterations focused on simplifying legends, improving pop-up organization, adjusting color schemes, and reorganizing navigation menus. This became especially important for ensuring the platform remained accessible to non-technical users.

Long-Term Maintainability

Maintainability became a recurring concern throughout the project. Due to many datasets having changing structures yearly, using inconsistent namings, depending on external APIs, and requiring manual downloading, the platform could not realistically rely on fully static pipelines. Instead, the architecture had to remain modular and adaptable to future changes. To partially address this issue, we therefore also included the development of a user manual, documentation of development and decisions, as well as technical documentation.

9. Critical Turning Points

Since we followed a Participatory Design approach, we continuously received feedback and made iterative changes throughout the project. However, there were also several critical turning points where our direction or plan had to change more significantly. The following section summarises these key turning points.

Machine Learning integration

We originally planned to include a Machine Learning (ML) tab to show trends alongside the map. While *Advocaat van de Aardet* thought it was a nice idea, there were concerns that people might treat the results as more certain than they actually are, which could be risky in a legal context. It was also agreed that our main focus should stay on improving the map itself.

Thus, we decided not to include ML. It became a clear trade-off between adding extra features and keeping the core tool strong and usable.

Key Changes

1. *ML was removed from website and only explored in final report*

Yearly Scale

One of the partner requirements was to include historical data. At first, we wanted to build a time slider so users could move between years and see how the data changed over time. However, this turned out to be more difficult than expected. The datasets were often too large to store locally, and in some cases there was no consistent historical data available, which could also lead to misleading comparisons.

We decided not to include the time-slider feature. Instead, we only added historical data for specific datasets, and provided a script for 010 Coding Collective, so they can expand it further in the future

Key Changes

1. *Removed time-slider feature*
2. *Added historical data for specific datasets, with a dropdown menu to choose what year to view*
3. *Provided a script for 010 Coding Collective for future development*

Homepage redesign

Early feedback revealed that users were not immediately able to understand the purpose of the platform or how it could support environmental litigation. While the map itself was intuitive, the broader project goals and intended audience were unclear. As a result, the homepage was redesigned to include a concise explanation of the tool, its objectives, and its relevance for environmental advocacy and legal action.

Key Changes

2. *Added a clear project introduction*
3. *Included more contextual information about the platform's purpose.*
4. *Informed users what each dataset contains*

Introduction of Icons and Information Features

As more datasets and functionalities were added, users expressed it was difficult to identify the different data sets if they were in the same format (for example hospitals and schools being both circle markers on the website). This led to the use of icons / shapes and clearer colour differences to make each data set easier to recognise while maintaining a clean design.

Key Changes

1. *Added information (i) icons next to datasets*
2. *Assigned specific icons to each dataset (e.g schools → graduation cap, KDR → animal symbols)*
3. *Improved and adjusted colour schemes to clearly separate different datasets*

Expansion of Datasets

One of the most significant findings from the focus groups was that several datasets considered essential for environmental litigation were missing or incorrectly implemented. Although our initial goal was to include eight core datasets, we were able to successfully integrate these and receive positive feedback from our partners. This then led to additional requests for further datasets, which shifted our focus from mainly refining the UI to implementing more environmental datasets.

Key Changes

1. *Replaced incorrectly implemented Woondeals (Administrative boundaries)*

2. *Added Natuurnetwerk Nederland (NNN)*
3. *Added cadastral parcel information*
4. *Added BAG verblijfslocaties*
5. *Added Waterschappen (water board boundaries)*

Sustainability and Long-Term Use

We knew from the start that we wanted to build something modular that could be maintained in the future, but how exactly to achieve this remained an open question. As development progressed, we had more discussions with *Advocaat van de Aarde* and the 010 Coding Collective, about hosting, maintenance, documentation, and future ownership. These topics were gradually clarified with our partners, and we agreed to stay in contact after the deadline to sort out any remaining questions and make the handover as smooth as possible.

Key Changes

1. *Developed technical documentations (including user guide, development and decisions log, and technical explanations)*
2. *Considered future maintenance and handover processes*
3. *Explored long-term hosting and organisational ownership options with 010 coding collective*

10. Lessons Learned

Perhaps the biggest lesson from the project was realizing how much environmental GIS work depends on infrastructure, preprocessing, and data management. Building an interactive environmental mapping platform required not only technical coding skills, but also understanding geospatial systems, datasets, legal workflows, usability, and additionally collaboration. The project also showed us that environmental data accessibility is not simply a visualization problem, but it's connected to data standardization, transparency, governance structures, and the ability of non-experts to meaningfully interpret complex spatial information.

Technical lessons

1. Working with Geospatial Data

One of the biggest lessons from the project was understanding how fundamentally different geospatial data is from traditional tabular datasets. Environmental GIS data involves not only attribute information, but also geometry, coordinate systems, spatial relationships, and rendering behavior. Throughout development, we learned the importance of:

- Understanding geometry types (points, lines, polygons, raster data)
- Choosing appropriate visualization methods depending on the dataset
- Structuring spatial data pipelines before beginning frontend development
- Separating preprocessing, storage, and visualization workflows

We also learned that integrating datasets is significantly more complex than simply displaying them together on a map. Spatial analysis requires careful consideration of scale, projection accuracy, geometry validity, and dataset limitations.

2. *Coordinate Systems & CRS Management*

Coordinate Reference Systems (CRS) became one of the most important technical concepts throughout the project. Early in development, we underestimated how critical CRS standardization is within GIS workflows. Most Dutch datasets use RD New (EPSG:28992), while web-based maps typically use WGS84 (EPSG:4326). We learned that inconsistent CRS handling can easily produce spatial inaccuracies, particularly when conducting distance calculations or buffer analysis. We learned the importance of converting datasets into one standardized CRS internally, and understanding reprojection before conducting analysis. It's also important to check CRS metadata carefully before ingestion, and recognize that some datasets may not define CRS information at all.

Through discussions with GIS experts, we also learned that storing and processing datasets in a single CRS is often more efficient and reliable than constantly converting between systems dynamically.

3. *Databases & Spatial Architecture*

We also learned the importance of designing scalable spatial data pipelines early on. Initially, we underestimated the complexity of storing and rendering large geospatial datasets. Working with millions of geometries demonstrated the importance of database chunking and batching and the limits of client-side rendering.

Collaboration Lessons

1. *Collaborative Development on GitHub*

Working collaboratively on a large GIS-based web application introduced several challenges working on GitHub and version control. Because multiple team members were frequently editing the same files simultaneously, merge conflicts became a recurring issue throughout development. This taught us the importance of smaller and more frequent commits, as well as having a clearer branch management. We also learned that better communication before editing shared files, and reviewing changes before merging, are also very crucial. Although these issues occasionally slowed development, they also improved our understanding of real collaborative software engineering workflows.

2. *Remote Infrastructure and Tailscale*

Using shared infrastructure such as PostgreSQL databases connected through Tailscale introduced additional collaboration challenges. We occasionally faced database connection issues, authentication problems, and inconsistent local environments. At the same time, this experience provided valuable exposure to collaborative backend infrastructure and remote database management.

Software-specific insights

1. QGIS

QGIS became an important learning tool throughout the project, particularly during the early exploration and preprocessing stages. We learned that QGIS is extremely useful for exploring spatial relationships visually, inspecting geometry and CRS information, as well as cleaning and exporting geospatial files. At the same time, we also learned its limitations for scalable web applications. While QGIS is great for analysis and preprocessing, much of the project's automation, ingestion, and frontend functionality ultimately required Python-based pipelines and web technologies.

2. Leaflet

Leaflet proved highly effective for building an accessible and interactive web-based GIS interface. Its simplicity and flexibility made it well-suited for rapid frontend prototyping and layer interaction. However, we also noticed several limitations. Rendering large polygon datasets can quickly become slow, and many layers require simplification or tiling. Complex spatial analysis is difficult client-side

11. Reflection on Final Product

What Went Well

The final prototype successfully demonstrates how environmental datasets can be brought together into a single, accessible platform. Throughout the focus groups, participants consistently described the interface as intuitive, visually appealing, and that it was easier to use than GIS (*honestly, every platform in the world is*). The integration of multiple environmental datasets into one interactive map reduces the need for users to hunt for data across numerous separate sources. The export functionality and buffer analysis further increased the value of the tool by allowing users to not only visually explore relationships but move towards evidence gathering and preliminary analysis.

Strengths

- 1. User-friendly and intuitive interface*
- 2. Integration of multiple environmental datasets in one platform*
- 3. Interactive map visualisation*
- 4. Buffer analysis for spatial investigations*
- 5. PDF and Excel export functionality*
- 6. Positive feedback from environmental lawyers and experts*

Even Better If

Despite these strengths, the final prototype remains limited in several ways. Many datasets were added later in the development process, which meant there was less time to refine both the tool and the newly integrated data. Furthermore, the tool has not been tested with broader user groups such as citizens, municipalities, or water authorities, despite its aim to support

public participation. The focus groups were conducted with a small number of highly specialised participants and therefore cannot be used to draw strong conclusions about general usability. In addition, the tool has not yet been evaluated against formal evidentiary requirements within Dutch legal proceedings.

Limitations

1. *Limited user testing sample*
2. *Reliance on partial or recently integrated datasets*
3. *No formal assessment of legal admissibility*
4. *Limited testing with non-expert users*
5. *Uncertainty regarding long-term hosting and maintenance*

Things We Wish We Had Done Differently

Looking back, one of our main mistakes was not working with real data earlier in the project. Because we relied on mock or incomplete datasets in the early stages, many of our discussions about the analytical value of the tool stayed quite theoretical for too long. This meant that certain limitations only became visible later in development, when the real datasets were finally integrated and tested. As a result, some design decisions had to be revised under time pressure, and we lost opportunities to iterate more effectively on functionality and data structure from the start. Additionally, we do think Option 2 (GeoJSON + turf.js + deck.gl) could have resulted in a more powerful tool. In hindsight, it might have been worth exploring it further or at least trying it out, even if only in a small prototype, to better understand what it could offer.

In hindsight

1. *Use real datasets from the beginning*
2. *Conduct more focus groups with more diverse users*
3. *Develop hosting plans earlier*
4. *Create a common data workflow, with naming and formatting, from the start to avoid later inconsistencies*
5. *Try out Option 2 (GeoJSON + turf.js + deck.gl) in a small prototype to test whether it could have been applied in the project*
6. *Focus on accessibility features (such as audio support for hearing-impaired users or adjustable font size for readability)*

Things we wish we included but did not

Due to time, technical, and resource constraints, several features were identified but could not be fully implemented within the scope of the Capstone project.

In a perfect world, we would have loved to included the features listed below:

Desired Additions

1. *More advanced analytical reporting functions beyond basic export outputs*
 - a. *For example, a template which produces dataset summaries, key features, legal frameworks, and data on the map as one downloadable document*
2. *Full Dutch-English translation options*

- a. *Potentially more languages too*
- 3. *Ai-chat box for help navigating the website*
 - a. *We know, AI is everywhere nowadays, but this could really be helpful for users who have no real idea about the datasets who are not familiar with the datasets and therefore need more explanation than what is immediately available on the website.*
- 4. *Shared mapping spaces where lawyers, scientists, NGOs and citizens can annotate and discuss the same map*
 - a. *Almost like a reddit thread. Different experts can share opinions, and non-experts can ask questions and learn more*
- 5. *Alert system notifying users when new relevant datasets are published in monitored areas*
 - a. *This could work as simple email notifications, where users can optionally leave their email to get updates*
 - b. *Alternatively, users could create an account to save their progress, views, and areas they are tracking, so they can easily come back to them later*

Future Outlook

The project shows how environmental justice goals can be combined with accessible geospatial tools in a meaningful way. At the moment, the prototype mainly works as a decision-support and exploratory mapping tool, but there is clear potential for it to grow into a more complete platform for environmental monitoring, advocacy, and legal support. The strong interest from environmental lawyers and scientific experts also suggests that there is a genuine need for a tool like this within the Dutch environmental sector.

What this tool could look like in the future

In the future, the platform could grow into a larger environmental system that brings together environmental, health, administrative, and legal datasets from the Dutch government in one place. With features like automated reporting, stronger spatial analysis, and tools for generating evidence, it could better support environmental organisations and legal practitioners in their work. If it were connected to real-time data sources and supported by long-term hosting, the platform could move beyond a university prototype and become a practical tool for environmental governance and public participation across the Netherlands.

Future possibilities

(Please refer to desired additions first, but if they seem too ambitious (the sky is the limit :)), here is a list of more realistic goals)

- 1. *Creation of full legal evidence reports including maps, selected layers, buffer outputs, metadata, timestamps, and methodological documentation structured for Dutch administrative court use*
- 2. *Real-time environmental monitoring using live datasets (e.g. air quality sensors, water quality measurements, emission data)*

3. *Advanced spatial and temporal analysis, such as year-to-year comparison, trend detection, and distance-based calculations (e.g. pollution source to school or Natura 2000 areas)*
4. *Integration of ML analysis (current models from the final report, with potential for additional methods and deeper analysis)*
5. *Integration with government open-data platforms (e.g. PDOK, DUO and other Dutch data websites) so datasets can be updated automatically, stay more consistent, and users can then find and access our platform more easily*
6. *Integrate accessibility features*
 - a. *Adjustable font sizes and scalable UI elements for better readability*
 - b. *Colourblind-friendly colour palette for map layers and data visualisations*
 - c. *Text alternatives for all icons and visual elements*
 - d. *Audio descriptions or summaries of key map insights for users with visual or reading difficulties*